

# Self-Contained Better-Notation Rewrite of the Schlumprecht Szlenk-Radius Converse

Human-requested companion note

**Important note.** This document is not the original LLM-generated output. It is a human-requested rewrite, requested by Antonio Acuaviva, of the proposed argument with improved notation and with the details made self-contained. The mathematical content is intended to be the same as in the original solution packet, but the notation has been reorganized to make the proof easier to check.

## 1 Setup

Let

$$\varphi(n) = \log_2(n+1) \quad (n \in \mathbb{N}).$$

Schlumprecht's space  $\mathcal{S}$  is the completion of  $c_{00}$  under the norm

$$\|x\|_{\mathcal{S}} = \max \left\{ \|x\|_{\infty}, \sup_{n \geq 2, E_1 < \dots < E_n} \frac{1}{\varphi(n)} \sum_{i=1}^n \|E_i x\|_{\mathcal{S}} \right\}.$$

Here  $E x$  denotes the coordinate restriction of  $x$  to the finite set  $E$ , and  $E_1 < \dots < E_n$  means the sets are successive.

We use the standard facts that the canonical basis of  $\mathcal{S}$  is 1-unconditional and shrinking, and that  $\mathcal{S}$  is reflexive. Thus  $e_N \rightarrow 0$  weakly in  $\mathcal{S}$ .

For a weak-star compact set  $K \subset X^*$ , the Szlenk derivation is

$$s_{\varepsilon}(K) = \{x^* \in K : \text{diam}(K \cap V) > \varepsilon \text{ for every weak-star neighbourhood } V \ni x^*\}.$$

In the present case  $X = \mathcal{S}^*$ , so  $X^* = \mathcal{S}$ . The corresponding inner Szlenk radius is

$$r_{\mathcal{S}^*}(\varepsilon) = \sup\{r > 0 : rB_{\mathcal{S}} \subset s_{\varepsilon}(B_{\mathcal{S}})\}.$$

For  $0 < \alpha < 1$ , define

$$a_{\alpha} = \inf_{n \in \mathbb{N}} \frac{\varphi(n+1) - \alpha}{\varphi(n)}.$$

The known upper bound from the source paper is

$$r_{\mathcal{S}^*}(\varepsilon) \leq a_{\varepsilon/2}.$$

The goal of the proposed argument is to prove the reverse inequality.

**Theorem.** For  $0 < \varepsilon < 2$ ,

$$r_{\mathcal{S}^*}(\varepsilon) = \inf_{n \in \mathbb{N}} \frac{\varphi(n+1) - \varepsilon/2}{\varphi(n)}.$$

## 2 Logarithmic notation

For  $i \geq 1$ , write

$$B_i = \varphi(i+1), \quad \Delta_i = \varphi(i+1) - \varphi(i).$$

Thus

$$B_i = \log_2(i+2), \quad \Delta_i = \log_2 \left( 1 + \frac{1}{i+1} \right).$$

This notation is kept throughout the proof. In particular, when the product lemma is applied later with  $i = k-1$  and  $j = m$ , the relevant quantities remain visibly

$$\Delta_{k-1}, \quad \Delta_m, \quad B_\ell, \quad \Delta_\ell.$$

## 3 The product lemma

We first record the elementary monotonicity fact behind the product lemma.

**Lemma.** *The function*

$$h(u) = \frac{u \log u}{u-1} \quad (u > 1)$$

*is increasing.*

*Proof.* Differentiate:

$$h'(u) = \frac{(u-1)(\log u + 1) - u \log u}{(u-1)^2} = \frac{u-1-\log u}{(u-1)^2}.$$

Since  $\log u \leq u-1$ , we have  $h'(u) \geq 0$ . □

**Lemma** (Product lemma). *For  $i, j \geq 1$ , put*

$$\ell = (i+1)(j+1) - 1.$$

*Then*

$$B_\ell \leq B_i B_j, \quad \Delta_\ell \leq \Delta_i \Delta_j.$$

*Proof.* The first inequality is equivalent to

$$\log_2(pq+1) \leq \log_2(p+1) \log_2(q+1) \quad (p, q \geq 2),$$

where  $p = i+1$  and  $q = j+1$ . For fixed  $p$ , consider

$$F_p(q) = \frac{\log(pq+1)}{\log(q+1)}.$$

A quotient-rule calculation gives

$$F'_p(q) = \frac{p(q+1) \log(q+1) - (pq+1) \log(pq+1)}{(pq+1)(q+1)(\log(q+1))^2}.$$

The numerator is non-positive because it is the comparison

$$\frac{(q+1) \log(q+1)}{q} \leq \frac{(pq+1) \log(pq+1)}{pq},$$

which follows from the monotonicity of  $h$ . Hence  $F_p$  is decreasing, so  $F_p(q) \leq F_p(1)$ . This gives

$$\frac{\log(pq+1)}{\log(q+1)} \leq \frac{\log(p+1)}{\log 2},$$

which is the first inequality.

For the increment inequality, put  $a = 1/p$  and  $b = 1/q$ . It is enough to prove

$$\log_2(1 + ab) \leq \log_2(1 + a) \log_2(1 + b) \quad (0 < a, b \leq 1).$$

For fixed  $a$ , define

$$G_a(b) = \frac{\log(1 + ab)}{\log(1 + b)}.$$

Again differentiating,

$$G'_a(b) = \frac{a(1 + b) \log(1 + b) - (1 + ab) \log(1 + ab)}{(1 + ab)(1 + b)(\log(1 + b))^2}.$$

Let  $u = 1 + b$  and  $v = 1 + ab$ . Since  $0 < a \leq 1$ ,  $v \leq u$ , and

$$a(u - 1) = v - 1.$$

The numerator is

$$au \log u - v \log v = (v - 1) \left( \frac{u \log u}{u - 1} - \frac{v \log v}{v - 1} \right) \geq 0$$

by the monotonicity of  $h$ . Hence  $G_a$  is increasing, so

$$G_a(b) \leq G_a(1) = \frac{\log(1 + a)}{\log 2}.$$

This proves the increment inequality. □

## 4 The scalar majorant

First note that  $0 < a_\alpha < 1$ . For positivity, since  $\varphi(n + 1) > \varphi(n)$  and  $\varphi(n) \geq 1$ ,

$$\frac{\varphi(n + 1) - \alpha}{\varphi(n)} > \frac{\varphi(n) - \alpha}{\varphi(n)} = 1 - \frac{\alpha}{\varphi(n)} \geq 1 - \alpha > 0.$$

Thus  $a_\alpha \geq 1 - \alpha > 0$ . For the upper bound, use  $\varphi(n + 1) - \varphi(n) \rightarrow 0$ . Choose  $n$  with

$$\varphi(n + 1) - \varphi(n) < \alpha.$$

Then

$$\frac{\varphi(n + 1) - \alpha}{\varphi(n)} < 1,$$

so  $a_\alpha < 1$ .

From the definition of the infimum, for every  $n \in \mathbb{N}$ ,

$$a_\alpha \leq \frac{\varphi(n + 1) - \alpha}{\varphi(n)}.$$

Equivalently,

$$\varphi(n)a_\alpha + \alpha \leq \varphi(n + 1).$$

For  $u \geq 0$ , define

$$M_\alpha(u) = \max \left\{ u, \alpha, \sup_{m \in \mathbb{N}} \frac{\varphi(m)u + \alpha}{\varphi(m + 1)} \right\}.$$

This scalar function is designed to dominate the cost of adding one far coordinate of size  $\alpha$ .

If  $0 \leq u < a_\alpha$ , then  $M_\alpha(u) \leq 1$ . Indeed,  $u < 1$  and  $\alpha < 1$ , while for every  $m$ ,

$$\frac{\varphi(m)u + \alpha}{\varphi(m + 1)} \leq \frac{\varphi(m)a_\alpha + \alpha}{\varphi(m + 1)} \leq 1.$$

## 5 Strengthened scalar estimate

**Lemma** (Strengthened scalar estimate). *Let  $0 < \alpha < 1$ ,  $0 \leq s \leq t$ , and  $k \geq 2$ . Then*

$$M_\alpha(s) + \min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \varphi(k)M_\alpha(t).$$

*Proof.* We prove the estimate separately for each type of term appearing in  $M_\alpha(s)$ .

First consider the term  $s$ . Since

$$\min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \varphi(k)t - s,$$

we have

$$s + \min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \varphi(k)t \leq \varphi(k)M_\alpha(t).$$

Next consider the term  $\alpha$ . Since

$$\min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \varphi(k-1)t,$$

we get

$$\alpha + \min\{\varphi(k-1)t, \varphi(k)t - s\} \leq \alpha + \varphi(k-1)t.$$

But  $M_\alpha(t)$  contains the affine term indexed by  $k-1$ :

$$M_\alpha(t) \geq \frac{\varphi(k-1)t + \alpha}{\varphi(k)}.$$

Thus

$$\alpha + \varphi(k-1)t \leq \varphi(k)M_\alpha(t).$$

It remains to consider an affine term indexed by  $m$ :

$$\frac{\varphi(m)s + \alpha}{\varphi(m+1)}.$$

For fixed  $t$ , put

$$F_m(s) = \frac{\varphi(m)s + \alpha}{\varphi(m+1)} + \min\{\varphi(k-1)t, \varphi(k)t - s\}.$$

The two quantities inside the minimum are equal when

$$\varphi(k-1)t = \varphi(k)t - s,$$

that is, when

$$s = (\varphi(k) - \varphi(k-1))t = \Delta_{k-1}t.$$

On  $0 \leq s \leq \Delta_{k-1}t$ , the minimum is  $\varphi(k-1)t$ , so  $F_m$  is increasing. On  $\Delta_{k-1}t \leq s \leq t$ , the minimum is  $\varphi(k)t - s$ , and the slope of  $F_m$  is

$$\frac{\varphi(m)}{\varphi(m+1)} - 1 < 0.$$

Hence  $F_m$  is maximized at  $s = \Delta_{k-1}t$ .

At that point,

$$F_m(\Delta_{k-1}t) = \frac{\varphi(m)\Delta_{k-1}t + \alpha}{\varphi(m+1)} + \varphi(k-1)t.$$

Putting this over the denominator  $\varphi(m+1)$ , we get

$$F_m(\Delta_{k-1}t) = \frac{\alpha + t(\varphi(m)\Delta_{k-1} + \varphi(k-1)\varphi(m+1))}{\varphi(m+1)}.$$

Using

$$\varphi(k) = \varphi(k-1) + \Delta_{k-1}, \quad \varphi(m+1) = \varphi(m) + \Delta_m,$$

one obtains

$$\varphi(m)\Delta_{k-1} + \varphi(k-1)\varphi(m+1) = \varphi(k)\varphi(m+1) - \Delta_{k-1}\Delta_m.$$

Therefore

$$F_m(\Delta_{k-1}t) = \frac{\alpha + t(\varphi(k)\varphi(m+1) - \Delta_{k-1}\Delta_m)}{\varphi(m+1)}.$$

It is enough to prove

$$M_\alpha(t) \geq t + \frac{\alpha - \Delta_{k-1}\Delta_m t}{\varphi(k)\varphi(m+1)}.$$

Apply the product lemma with  $i = k-1$  and  $j = m$ . Then

$$\ell = (k-1+1)(m+1) - 1 = k(m+1) - 1,$$

and

$$B_\ell \leq B_{k-1}B_m = \varphi(k)\varphi(m+1),$$

while

$$\Delta_\ell \leq \Delta_{k-1}\Delta_m.$$

The affine term indexed by  $\ell$  in  $M_\alpha(t)$  gives

$$M_\alpha(t) \geq \frac{\varphi(\ell)t + \alpha}{\varphi(\ell+1)}.$$

Since

$$B_\ell = \varphi(\ell+1), \quad \Delta_\ell = \varphi(\ell+1) - \varphi(\ell),$$

we have

$$\varphi(\ell) = B_\ell - \Delta_\ell.$$

Hence

$$M_\alpha(t) \geq t + \frac{\alpha - \Delta_\ell t}{B_\ell}.$$

If

$$\alpha - \Delta_{k-1}\Delta_m t < 0,$$

then

$$t + \frac{\alpha - \Delta_{k-1}\Delta_m t}{\varphi(k)\varphi(m+1)} < t,$$

and the desired inequality follows from  $M_\alpha(t) \geq t$ .

If

$$\alpha - \Delta_{k-1}\Delta_m t \geq 0,$$

then

$$\alpha - \Delta_\ell t \geq \alpha - \Delta_{k-1}\Delta_m t \geq 0,$$

and

$$B_\ell \leq \varphi(k)\varphi(m+1).$$

Therefore

$$\frac{\alpha - \Delta_\ell t}{B_\ell} \geq \frac{\alpha - \Delta_{k-1}\Delta_m t}{\varphi(k)\varphi(m+1)}.$$

Thus the affine-term case follows.

The estimate has now been checked for  $s$ , for  $\alpha$ , and for every affine term indexed by  $m$ . Hence it holds for  $M_\alpha(s)$ .  $\square$

## 6 A comparison lemma for the Schlumprecht norm

Define iterative norms on  $c_{00}$  by

$$\|x\|_0 = \|x\|_\infty,$$

and

$$\|x\|_{r+1} = \max \left\{ \|x\|_\infty, \sup_{n \geq 2, E_1 < \dots < E_n} \frac{1}{\varphi(n)} \sum_{i=1}^n \|E_i x\|_r \right\}.$$

Then

$$\|x\|_{\mathcal{S}} = \sup_{r \geq 0} \|x\|_r.$$

**Lemma** (Local comparison). *Let  $y \in c_{00}$ . Suppose there is a function*

$$\rho : [\mathbb{N}]^{<\omega} \rightarrow [0, \infty),$$

where

$$[\mathbb{N}]^{<\omega} = \{F \subset \mathbb{N} : F \text{ is finite}\},$$

such that, for every finite  $F \subset \mathbb{N}$ ,

$$\|Fy\|_\infty \leq \rho(F),$$

for every  $G \subset F$ ,

$$G \subset F \implies \rho(G) \leq \rho(F),$$

and, whenever  $k \geq 2$  and  $E_1 < \dots < E_k \subset F$ ,

$$\sum_{i=1}^k \rho(E_i) \leq \varphi(k) \rho(F).$$

Then

$$\|Fy\|_{\mathcal{S}} \leq \rho(F)$$

for every finite  $F \subset \mathbb{N}$ .

*Proof.* We prove by induction on  $r$  that

$$\|Fy\|_r \leq \rho(F)$$

for every finite  $F$ . The case  $r = 0$  is exactly the assumption  $\|Fy\|_\infty \leq \rho(F)$ .

Assume the claim holds for  $r$ . For  $E_1 < \dots < E_k \subset F$ ,

$$\sum_{i=1}^k \|E_i y\|_r \leq \sum_{i=1}^k \rho(E_i) \leq \varphi(k) \rho(F).$$

Together with  $\|Fy\|_\infty \leq \rho(F)$ , this gives

$$\|Fy\|_{r+1} \leq \rho(F).$$

Taking the supremum over  $r$  proves the result. □

## 7 Far-coordinate perturbation

**Proposition.** *Let  $0 < \alpha < 1$ , let  $x \in c_{00}$ , and assume*

$$\|x\|_{\mathcal{S}} < a_{\alpha}.$$

*If  $N > \max \text{supp } x$ , then*

$$\|x + \alpha e_N\|_{\mathcal{S}} \leq 1 \quad \text{and} \quad \|x - \alpha e_N\|_{\mathcal{S}} \leq 1.$$

*Proof.* By 1-unconditionality, it is enough to prove the estimate for  $|x| + \alpha e_N$ . Thus assume  $x \geq 0$ , and set

$$y = x + \alpha e_N.$$

For each finite  $F \subset \mathbb{N}$ , define

$$\rho(F) = \begin{cases} \|Fx\|_{\mathcal{S}}, & N \notin F, \\ M_{\alpha}(\|Fx\|_{\mathcal{S}}), & N \in F. \end{cases}$$

The lattice monotonicity of the Schlumprecht norm and the monotonicity of  $M_{\alpha}$  give

$$G \subset F \implies \rho(G) \leq \rho(F).$$

Also,

$$\|Fy\|_{\infty} \leq \rho(F),$$

because  $M_{\alpha}(u) \geq u$  and  $M_{\alpha}(u) \geq \alpha$ .

It remains to verify the local comparison inequality. Fix finite  $F \subset \mathbb{N}$ , and let

$$E_1 < \dots < E_k \subset F, \quad k \geq 2.$$

If  $N \notin F$ , then

$$\sum_{i=1}^k \rho(E_i) = \sum_{i=1}^k \|E_i x\|_{\mathcal{S}} \leq \varphi(k) \|Fx\|_{\mathcal{S}} = \varphi(k) \rho(F).$$

Assume now that  $N \in F$ . If none of the  $E_i$  contains  $N$ , then

$$\sum_{i=1}^k \rho(E_i) = \sum_{i=1}^k \|E_i x\|_{\mathcal{S}} \leq \varphi(k) \|Fx\|_{\mathcal{S}} \leq \varphi(k) M_{\alpha}(\|Fx\|_{\mathcal{S}}) = \varphi(k) \rho(F).$$

Finally suppose exactly one set  $E_j$  contains  $N$ , and fix this  $j$ . Then

$$\sum_{i=1}^k \rho(E_i) = M_{\alpha}(\|E_j x\|_{\mathcal{S}}) + \sum_{i \neq j} \|E_i x\|_{\mathcal{S}}.$$

The remaining pieces satisfy two estimates. First,

$$\sum_{i \neq j} \|E_i x\|_{\mathcal{S}} \leq \varphi(k-1) \|Fx\|_{\mathcal{S}}.$$

For  $k = 2$ , this is just monotonicity, since  $\varphi(1) = 1$ . For  $k > 2$ , it follows from the Schlumprecht recursive inequality applied to the  $k-1$  remaining blocks. Second, using the full  $k$ -tuple,

$$\|E_j x\|_{\mathcal{S}} + \sum_{i \neq j} \|E_i x\|_{\mathcal{S}} \leq \varphi(k) \|Fx\|_{\mathcal{S}},$$

so

$$\sum_{i \neq j} \|E_i x\|_{\mathcal{S}} \leq \varphi(k) \|Fx\|_{\mathcal{S}} - \|E_j x\|_{\mathcal{S}}.$$

Therefore

$$\sum_{i \neq j} \|E_i x\|_{\mathcal{S}} \leq \min\{\varphi(k-1) \|Fx\|_{\mathcal{S}}, \varphi(k) \|Fx\|_{\mathcal{S}} - \|E_j x\|_{\mathcal{S}}\}.$$

By the strengthened scalar estimate,

$$\begin{aligned} \sum_{i=1}^k \rho(E_i) &\leq M_{\alpha}(\|E_j x\|_{\mathcal{S}}) + \min\{\varphi(k-1) \|Fx\|_{\mathcal{S}}, \varphi(k) \|Fx\|_{\mathcal{S}} - \|E_j x\|_{\mathcal{S}}\} \\ &\leq \varphi(k) M_{\alpha}(\|Fx\|_{\mathcal{S}}) = \varphi(k) \rho(F). \end{aligned}$$

The local comparison lemma gives

$$\|Fy\|_{\mathcal{S}} \leq \rho(F)$$

for every finite  $F$ . Taking  $F = \text{supp } x \cup \{N\}$ , we get

$$\|x + \alpha e_N\|_{\mathcal{S}} \leq M_{\alpha}(\|x\|_{\mathcal{S}}) \leq 1,$$

because  $\|x\|_{\mathcal{S}} < a_{\alpha}$ . The estimate for  $x - \alpha e_N$  follows by unconditionality.  $\square$

## 8 The reverse Szlenk-radius inclusion

Let  $0 < \varepsilon < 2$ , and put

$$\beta = \varepsilon/2.$$

We prove

$$r_{\mathcal{S}^*}(\varepsilon) \geq a_{\beta}.$$

First take  $x \in c_{00}$  with

$$\|x\|_{\mathcal{S}} < a_{\beta}.$$

Choose  $\alpha$  such that

$$\beta < \alpha < 1 \quad \text{and} \quad \|x\|_{\mathcal{S}} < a_{\alpha}.$$

This is possible because the map  $\alpha \mapsto a_{\alpha}$  is continuous: it is the infimum of affine functions

$$\alpha \mapsto \frac{\varphi(n+1) - \alpha}{\varphi(n)}$$

whose slopes have absolute value at most 1.

For every sufficiently large  $N$ , the far-coordinate perturbation result gives

$$x + \alpha e_N \in B_{\mathcal{S}}, \quad x - \alpha e_N \in B_{\mathcal{S}}.$$

Since  $e_N \rightarrow 0$  weakly, both points converge weakly to  $x$ . Their distance is

$$\|(x + \alpha e_N) - (x - \alpha e_N)\|_{\mathcal{S}} = \|2\alpha e_N\|_{\mathcal{S}} = 2\alpha > \varepsilon.$$

Therefore every weak neighbourhood of  $x$  meets  $B_{\mathcal{S}}$  in a set of diameter  $> \varepsilon$ . Hence

$$x \in s_{\varepsilon}(B_{\mathcal{S}}).$$

Now let  $r < a_{\beta}$ . If  $z \in rB_{\mathcal{S}}$ , choose finitely supported  $x_n \rightarrow z$  in norm with  $\|x_n\|_{\mathcal{S}} < a_{\beta}$  for all large  $n$ . By the finite-support case,

$$x_n \in s_{\varepsilon}(B_{\mathcal{S}})$$



eventually. The set  $s_\varepsilon(B_S)$  is weakly closed, hence norm closed, so

$$z \in s_\varepsilon(B_S).$$

Thus

$$rB_S \subset s_\varepsilon(B_S) \quad (r < a_\beta),$$

and therefore

$$r_{S^*}(\varepsilon) \geq a_\beta.$$

Together with the known upper bound, this gives

$$r_{S^*}(\varepsilon) = a_{\varepsilon/2} = \inf_{n \in \mathbb{N}} \frac{\varphi(n+1) - \varepsilon/2}{\varphi(n)}.$$

## Comment on notation

The original proof introduced several temporary letters for quantities such as  $\varphi(k)$ ,  $\varphi(m+1)$ ,  $\Delta_{k-1}$ ,  $\Delta_m$ ,  $B_\ell$ , and  $\Delta_\ell$ . This rewrite keeps the logarithmic weights and increments visible. In particular, the key use of the product lemma in the scalar estimate is simply

$$B_\ell \leq \varphi(k)\varphi(m+1), \quad \Delta_\ell \leq \Delta_{k-1}\Delta_m,$$

where

$$\ell = k(m+1) - 1.$$